to follow the progress of the substance throughout all its reactions, and all we need know is the half-life of that radioactive isotope to determine the quantity of it present at any given moment. The measurement of the intensity of radioactivity gives a result no less true than that which we should get by weighing the substance itself—in fact, as the above example indicates, an even more reliable result in many an instance, since it will show infallibly whether or not the substance in question is actually the one sought.

Secondly, in chemistry it may be necessary at times to investigate exchange processes, which used to be inaccessible to every method of investigation, particularly where it is a matter of an interchange of elements having the same properties between substances. The question to be answered may be, for instance, whether the sulphur atoms in sulphuric acid ions and sulphurous acid ions are interchanged between these two substances when they combine. Up to now, the difficulty consisted in the fact that it was impossible to distinguish the atoms of one of these two substances from the atoms of the other. But today, radioactivity affords the possibility of labelling at least some —but in any case a sufficient number—of the atoms of one of the two substances. If after the subsequent separation of the two substances, the atoms thus labelled are found to be present in the other substance, this will prove that atoms have been interchanged. Such experiments have shown that an interchange of sulphur atoms actually occurs between the SO''_4 and SO''_3 ions. (The two apostrophes after the symbol indicate the double negative charge of these ions.)

There are some further chemical applications I wish to mention here very briefly. The following table gives some idea of the various possibilities of application. Let us begin to consider inorganic chemistry and start with the study of new chemical elements.

There are certain elements, the places of which in the table of the periodic system were vacant till quite recently. These are elements which—to use the customary mode of expression—had not yet been discovered but were assumed to occur in nature. The best known among these are the elements of the atomic numbers (in other words, nuclear charges) 43 and 61. It was long believed that element 43 had been discovered in nature, and it was named *masurium.* But today we have every reascn to believe that this was an error, and that no stable element of this kind can exist. For in the meantime every isotope of this element of appreciable stability has been produced artificially, and every one of them has been proved to be radioactive. The mass number of the longest-lived isotope is 99; its half-life is approximately four million years. Since this is a short period of time relative to the age of the earth, element 43 cannot possibly occur in nature in any measurable quantity. Since this element can be produced artificially in the uranium reactor, it has recently been renamed *technetium* (symbol: *Tc*). The above mentioned most stable